

Nicholas Coles

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Education

University of Michigan

Ann Arbor, MI

Bachelor of Science in Computer Science

2021 - 2025

- Relevant Courses: Data Structures & Algorithms, Intro Artificial Intelligence, Information Retrieval, Software Engineering, ML in Earth & Climate, Multidisciplinary Software Development
- Activities: VP of Professional Development, Zeta Pi Tech. Frat., Team Lead at Ethical Investments Club

Experience

J.P. Morgan Chase

Chicago, IL

Software Engineer I

Jul 2025-Present

- Sole engineer for an AI portfolio tool covering 600 projects; turned a hard-coded prototype into a production application in one month using React/TypeScript, Python, PostgreSQL, Databricks, and LLM agents, replacing quarterly slide reporting with live project updates and reaching 5K first-month visits and ~100 contributors.
- Tested different source-context setups and few-shot examples with project owners to generate accurate executive summaries from existing portfolio data while requiring little additional input from users.
- Added agent and tool-call status tracking from Python through React, reducing latency complaints by 65% and response-quality complaints by 15% without changing the underlying model, based on weekly surveys sent to ~500 users with ~40% participation.
- Built a Jira pipeline that reduced 1–2 GB of project data into ~2 pages of useful context, cutting preparation from 60s to 5s; generated and stored 15 reports daily with optional refresh, reducing worst-case load time from 2 minutes to 1 second and LLM calls by up to 98%.
- Built a Python agent cookbook used by 90 engineers, connecting Jira, Confluence, and Splunk and adding configurable workflows, search limits, dependency discovery, API mocking, and usage tracking; led workshops and collected daily feedback from developers.

Software Engineering Intern

Dec 2023-Dec 2024

- Built an email-attachment extraction pipeline using Donut and OCR, automating 77% of service-agent data entry.
- Evaluated Mixtral, Llama, and DistilBERT for production classification and summarization, improving accuracy from 84% to 92%.
- Led a six-person subteam integrating LangChain and EasyOCR, expanding document ingestion beyond PDFs to multiple file formats.

Protiviti

Chicago, IL

Enterprise Data & Analytics Intern

May 2023-Aug 2023

- Automated regulatory test-result analysis for a Fortune 500 lender using Python and Pandas, reducing manager review from four hours to 30 seconds.
- Built a Tkinter interface that reduced task time by 50% and enabled nontechnical analysts to operate the automation independently.

Projects

- Built [TraceMark](#), a Python/FastAPI middleware that embeds HMAC/HKDF-derived fingerprints into LLM text for provider-agnostic attribution.
- Validated the detector across 518K+ real-world documents and up to 10,000 candidate fingerprints, testing false positives and author-style bias.
- Built adversarial benchmarks for robustness, capacity, and latency; achieved <100 ms processing at 1K words and quantified real-world watermarking limits.

Skills

Languages: Python, Java, TypeScript, JavaScript, SQL

Generative AI: LLM agents, tool calling, retrieval-augmented generation (RAG), prompt and context engineering, few-shot prompting, model routing, LLM experimentation

Frameworks & Data: React, LangChain, Spring, PostgreSQL, Databricks, Kafka, Cassandra, Pandas

Cloud & Tooling: AWS, REST APIs, Jenkins, CI/CD, Splunk, Git